

Improving depth-resolution, in-plane contrast, and reducing non-uniformity artifacts for wide-angle DBT

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Abstract.

Purpose: This work aims to develop an image reconstruction algorithm for wide-angle digital breast tomosynthesis (DBT) that has improved depth resolution and in-plane contrast while reducing non-uniformity artifacts.

Approach: The image reconstruction algorithm is an extension of our prior work on sparsity-regularized iterative image reconstruction. The algorithm is performed in two stages as explained in a prior work. The first stage consists of a low-resolution reconstruction that exploits sparsity for quantitative accuracy. In this work, this first stage is augmented with a formulation that includes the estimation of a “background” image, which absorbs low-frequency artifacts that cause image non-uniformity.

Results: The new algorithm is demonstrated on a patient case for which the data are acquired on a wide-angle DBT system.

Conclusion: The results on the shown case indicate that the algorithm design goals have been met, but additional empirical results and task-based assessment are needed to strengthen this conclusion.

Keywords: digital breast tomosynthesis, image reconstruction, sparsity regularization.

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1 Introduction

The impact of Professor Barrett’s work on our research is profound and multi-faceted and it would be impossible to detail every instance where his influence was felt. Here, in this special issue article, we focus specifically on our recent line of investigation into sparsity-exploiting image reconstruction for digital breast tomosynthesis (DBT). In this project, we have mainly made use of Chapter 7 “Deterministic Description of Imaging Systems” in the *Foundations of Image Science* (FIS),¹ written together with Dr. Kyle Myers. Understanding the details of this chapter is crucial to anyone who is interested in developing image reconstruction algorithms, because image reconstruction is essentially an inversion of an imaging system model and thus a clear understanding of the types of image system models is necessary.

The organization of the paper is as follows: Sec. 2 explains how the taxonomy of imaging system models is utilized to formulate our studies on iterative image reconstruction for limited-data systems such as DBT. Sec. 3 then develops an extension to our previous DBT work that addresses depth resolution, in-plane contrast, and non-uniformity artifacts; and Sec. 4 demonstrates the algorithm on a patient case acquired on a wide-angle DBT system.

2 Imaging system models and sparsity-exploiting image reconstruction in digital breast tomosynthesis

Chapter 7 in FIS describes three types imaging system models: continuous-to-continuous (CC), continuous-to-discrete (CD), and discrete-to-discrete (DD). For X-ray tomography, the X-ray and Radon transforms are instances of CC models, where the object model is a continuous function of spatial variables and the measurement space is imagined to be a continuous function of X-ray source angle and detector receptor location. The CD model, however, is a more accurate description for any form of digital imaging because digital detectors consist of discrete arrays of pixels. Finally, the DD model must be contended with because image processing or reconstruction algorithms are implemented on computers, which necessarily discretize images so that they can be stored in memory and manipulated by the various programmed processing units. In addition fully utilizing Dr. Barrett's taxonomy of imaging models, his emphasis on task-based image quality assessment is felt throughout all of our work on imaging.

For analytic image reconstruction, the filtered back-projection (FBP) formulas are derived to be the inverse of the CC divergent-ray or Radon transforms. Because the measured data are discrete, some form of interpolation is implicit when creating the input sinogram to an FBP algorithm. Because interpolation may not perfectly reconstruct the continuous sinogram, the output image

values will have error even for ideal, noiseless sinogram measurements. Furthermore, the FBP algorithm can only be executed on a computer for a finite spatial grid. Thus, interpolation is again needed if the display pixel grid does not match the grid of points, which has been reconstructed by FBP. For iterative image reconstruction, the DD model plays a larger role because the scanned object is represented by a finite set of expansion functions, which is most commonly taken to be a grid of voxels. Thus, another source of error is introduced; namely, the divergent-ray transform of a grid of voxels likely does not match the divergent-ray transform of a continuous object function for any set of voxel coefficients. Thus discretization error is an issue with which all iterative algorithms must contend.

Turning to sparsity-exploiting image reconstruction in X-ray tomography, we write the canonical optimization problem for compressive sensing (CS)²

$$\arg \min_f \|\nabla f\|_1 \text{ such that } Xf = g, \quad (1)$$

where f represents the image; g is the sinogram; X is the X-ray, or divergent-ray, transform; and the objective function is the image total variation (TV). This optimization is designed to recover the image f under the assumption that f has a sparse gradient and the sinogram data is ideal; namely the data are generated from the true underlying object f_0 , i.e. $g = Xf_0$. This CS optimization becomes interesting when the measured data coverage is less than what would be needed for FBP image reconstruction; i.e. the equation $Xf = g$ does not have a unique solution. As of yet, we have not specified the type of imaging model; this optimization can be expressed in CC, CD, and DD forms. It turns out that the vast majority of the work on CS does in fact assume a DD imaging system model; the conditions for recovery of f , discussed in the literature, mainly focus

on the number of samples in g being some factor larger than the number of non-zeros in the image gradient. Such conditions only make sense with a DD formulation. Accordingly, ∇ becomes a matrix encoding a finite differencing approximation to the continuous gradient; X is the discretized X-ray transform; g is a set of sinogram samples; and f is a vector of image basis coefficients, most commonly voxel values.

For X-ray tomography, the CS optimization problem was originally formulated in CT to address view-angle undersampling, limited angular-range scanning, and cone-beam artifacts from a circular scan,^{3,4} but as explained in Pan *et al.*⁵ CS can address a number of under-sampling issues such as projection truncation, bad detector pixels, and exterior/interior tomography. An important point to understand about all of these sampling challenges is that their degree of difficulty can be tuned for DD imaging system models. Because much CS work focuses on data sampling, mentioning the image basis expansion in passing or perhaps not at all, it leaves the impression that the sampling challenge is being addressed for CC or CD imaging system models and that a continuous image function – or a close approximation thereof – is being recovered. The reality is, however, that DD imaging system models are being employed, and only a finite set of image basis coefficients are being recovered. With this being the case any sampling challenge can be overcome simply by reducing the size of the basis expansion set. The price to be paid is that discretization error increases when performing image reconstruction on real data or simulation data generated with a continuous image model.

These issues surrounding DD modeling are central to our work on the limited angular-range image reconstruction problem that arises in DBT. Our first steps in accurately solving a DD model for DBT were reported in Zhang *et al.*,⁶ where the optimization problem of interest was a variant of TV-based image reconstruction that used directional TV (DTV) constraints. Following up on

this work, a thorough investigation on wide-angle DBT image reconstruction with DTV penalties was presented in Sidky *et al.*⁷ The wide-angle system studies was that of the Siemens Mammomat scanner that has 85 micron detector pixel resolution and scans the breast with 25 shots covering a 50 degree scan arc. The first study in that paper directly addressed the issue of "tuning" the DD model in order to get accurate image reconstruction for wide-angle DBT. Based on those studies, it was determined that relatively accurate and stable image reconstruction could be performed in a volume composed of cubic voxels with a width four times the width of the detector pixels. The remainder of that work investigated the impact of discretization error when the data are generated from a CD breast phantom model, and additionally the DTV algorithm was applied to DBT image reconstruction for a physical phantom obtained on a Mammomat scanner.

Central to all of these studies is the need for accurate solvers for non-smooth convex optimization problems. Many of the DD imaging system models of interest do not have analytic solutions available and they involve convex optimization problems with non-smooth terms such as the DTV norm or hard constraints such as non-negativity. The only way to provide evidence that the DD imaging system model of interest can be solved is to gather empirical results with computer-simulated phantoms, where the ground truth is known exactly. It must then be demonstrated that the phantom can be recovered perfectly from ideal noiseless data generated with this phantom. For this reason, we have been consistently adapting and refining the primal-dual hybrid gradient (PDHG) algorithm⁸ for X-ray tomographic image reconstruction,⁹ where the latest development is focused on efficient implementation for multi-term optimization problems, which are specifically needed for DD imaging system models relevant to DBT.¹⁰

The imaging system model design in this latest work is heavily influenced by the ideas discussed in FIS. The overall design is to split the reconstruction problem into two stages: a low-

resolution sparsity-exploiting reconstruction using a least-squares data fidelity with DTV regularization (lowres-LSQ-DTV), and high resolution Tikhonov-regularized least-squares (highres-LSQ-Tik). The design goal of lowres-LSQ-DTV is to minimize image root-mean-square-error (RMSE) at low resolution. The low-resolution aspect is needed because "inverse crime" inversion of the DD imaging model tells us that accurate image reconstruction becomes possible for cubic voxels four times larger than the detector pixels for wide-angle DBT as demonstrated in Refs. 7, 10. To avoid artifacts at this low-resolution both the object model and input data make use of gaussian smoothing that helps to avoid a jagged image representation and to ensure that the data fed into lowres-LSQ-DTV does not contain frequency information that is not representable with the low-resolution image grid; the need for this data filtering becomes clear from the various forms of discretization error explained in Chapter 7 of FIS. The use of large voxels and the fact that lowres-LSQ-DTV is optimized on image RMSE means that small, yet important, features in the image could be lost due to over-regularization – as demonstrated by a signal detection study using the channelized Hotelling observer machinery that was so well-explained in FIS.¹¹ To bring in the necessary fine details into the final image, the lowres-LSQ-DTV output image is upsampled to f_{DTV} and this image initializes the highres-LSQ-Tik algorithm, where the Tikhonov penalty terms also ensures proximity to f_{DTV} . Because lowres-LSQ-DTV provides a low RMSE initial image, the parameters of highres-LSQ-Tik can then be objectively determined based solely on clinical tasks of interest to DBT such as the detection of spiculations or microcalcifications using the machinery developed in FIS.

3 Coupled lowres-LSQ-DTV image reconstruction for estimating background artifacts

In this section, we present another advance of our DBT image reconstruction framework that deals with low-frequency background artifacts. Such artifacts enter into the raw reconstruction for a variety of reasons: imperfect knowledge of the X-ray illumination in each projection, X-ray scatter, and blurred high-density regions above or below the viewing plane. These low-frequency artifacts complicate display of the in-plane slices because they limit the ability to set the gray scale window for maximizing contrast of features of interest. In our previous work, Ref. 7 reconstructed in-plane slices were post-processed to remove low-frequency artifacts by fitting a low-order 2D polynomial to each slice and dividing the raw slice image by this fit. While effective for image presentation, it is not a 3D solution and it loses quantitative information.

Here, we augment lowres-LSQ-DTV, developed in Ref. 10, to coupled lowres-LSQ-DTV where three images are reconstructed simultaneously. The raw reconstructed image is denoted by f_1 ; the “background” image, which will primarily contain the low-frequency artifacts, is denoted by f_2 ; and the difference image is $f_3 = f_1 - f_2$. The optimization problem for coupled lowres-LSQ-DTV is

$$\begin{aligned}
& \arg \min_{f_1, f_2, f_3} \left\{ \alpha_x \|\partial_x f_1\|_1 + \alpha_y \|\partial_y f_1\|_1 + \alpha_a \|\partial_a f_1\|_1 + \alpha_b \|\partial_b f_1\|_1 + \alpha_1 \|f_1\|_1 \right. \\
& \quad + \alpha_x \|\partial_x f_2\|_1 + \alpha_y \|\partial_y f_2\|_1 + \alpha_a \|\partial_a f_2\|_1 + \alpha_b \|\partial_b f_2\|_1 + \alpha_1 \|f_2\|_1 \\
& \quad \left. + \alpha_3 \|I f_3\|_1 \quad \text{such that} \quad f_1 - f_2 = f_3, \right. \\
& \quad \left. \|R[c](XG[d_1]f_1 - g)\|_2 \leq \epsilon_1 \cdot \sqrt{N_{\text{data}}}, \text{ and} \right. \\
& \quad \left. \|R[c](XG[d_2]f_2 - G_{\text{det}}[d_d]g)\|_2 \leq \epsilon_2 \cdot \sqrt{N_{\text{data}}} \right\}, \tag{2}
\end{aligned}$$

where g represents the DBT projection data and N_{data} is the size of this dataset; X is the DBT X-ray transform that computes a projection data estimate based on an image estimate f ; the operator $G[d]$ performs 3D gaussian blurring with width $d = (d_x, d_y, d_z)$ and likewise $G_{\text{det}}[d]$ performs 2D gaussian blurring on each projections with width $d = (d_u, d_v)$; the operator $R[c]$ performs projection filtering with the square root of the ramp filter with frequency cutoff parameter c ; the parameters ϵ_1 and ϵ_2 are RMSE constraint parameters on the projection estimate for images f_1 and f_2 ; the objective function consists of a number of sparsity-inducing ℓ_1 -norm terms; the partial derivative symbol represents the finite differencing approximation to the derivative in the direction indicated in the subscript, where a and b represent oblique directions corresponding to vectors perpendicular to first and last X-ray source position in the xz -plane, see Sec. 6; the parameters α_x , α_y , α_a , α_b , and α_1 specify the relative strength of each sparsity-inducing term by normalizing the sum of these parameters to 1; note that the same parameters are used for f_1 and f_2 ; and finally, the parameter α_3 sets the strength on the ℓ_1 penalty for the difference image f_3 . The PDHG algorithm for this optimization is derived by mapping it onto the generic PDHG form; Sec 6 shows how this mapping is done and it specifies the step-size parameterization.

A large part of the image model design leading to Eq. (2) is explained in our previous paper Ref. 10. The new aspect is setting up the model for simultaneous optimization of the raw image f_1 and the “background” image f_2 and coupling these two images through the ℓ_1 -norm penalty on f_3 . Use of two data fidelity terms with low- and high-frequency bands has the potential to improve algorithm convergence¹² in addition to the estimation of f_1 and f_2 images. The use of the ℓ_1 -norm penalty on f_3 is to encourage small differences between f_1 and f_2 to continue to decrease, and more importantly, the ℓ_1 -norm does not strongly penalize large differences. We are expecting that f_1 will deviate from f_2 for in-focus features of the breast. For the data fidelity term involving f_2 , the

intention for introducing the gaussian blur $G[d_2]$ is to smooth out in-plane structures; accordingly the x and y components of the width vector d_2 should be large, on the order of centimeters, while the depth or z component of d_2 should be on the order of the slice thickness. The corresponding blur for projection data, $G_{\text{det}}[d_d]$, should be slightly larger than the in-plane blurring components of d_2 in order to suppress frequency components that would pick up real structures in the breast and so that filtered data can be consistent with the designed imaging model for f_2 . For the data fidelity term involving f_1 , the purpose of the gaussian blur $G[d_1]$ is to smooth the blockiness introduced by the use of large voxels before X-ray projection.

4 Results on wide-angle DBT image reconstruction for a patient case

We demonstrate the application of coupled lowres-DTV-LSQ on a patient case acquired on a Siemens Mammomat B.brilliant scanner. Institutional Review Board (IRB) approval was obtained for this retrospective, Health Insurance Portability and Accountability Act-compliant study conducted from data found in institutional patient databases. Written informed consent was waived by the IRB. The scan configuration involves 25 X-ray projections covering an arc of 50 degrees, which is currently the widest angle DBT on the market. The X-ray detector is 3584×2816 pixels with each pixel being 85 microns squared. The long-axis of the detector u is parallel to the direction of the scan and the x -axis of the image volume, while the short-axis v is parallel to the y -axis of the image volume. To process the transmission data to projection data, the air scan fluence, $I_0(u, v)$, is estimated from a strip on the detector between pixels 2000 and 2500 in the v -direction, because these detector rows are well beyond the breast projection. The u dependent fluence, $\bar{I}_0(u)$, is estimated by averaging $I_0(u, v)$ over v between 2000 and 2500. The estimated projection data, $g(u, v)$, are obtained by standard negative logarithm processing; i.e. $g(u, v) = -\ln(I(u, v)/\bar{I}_0(u))$. The

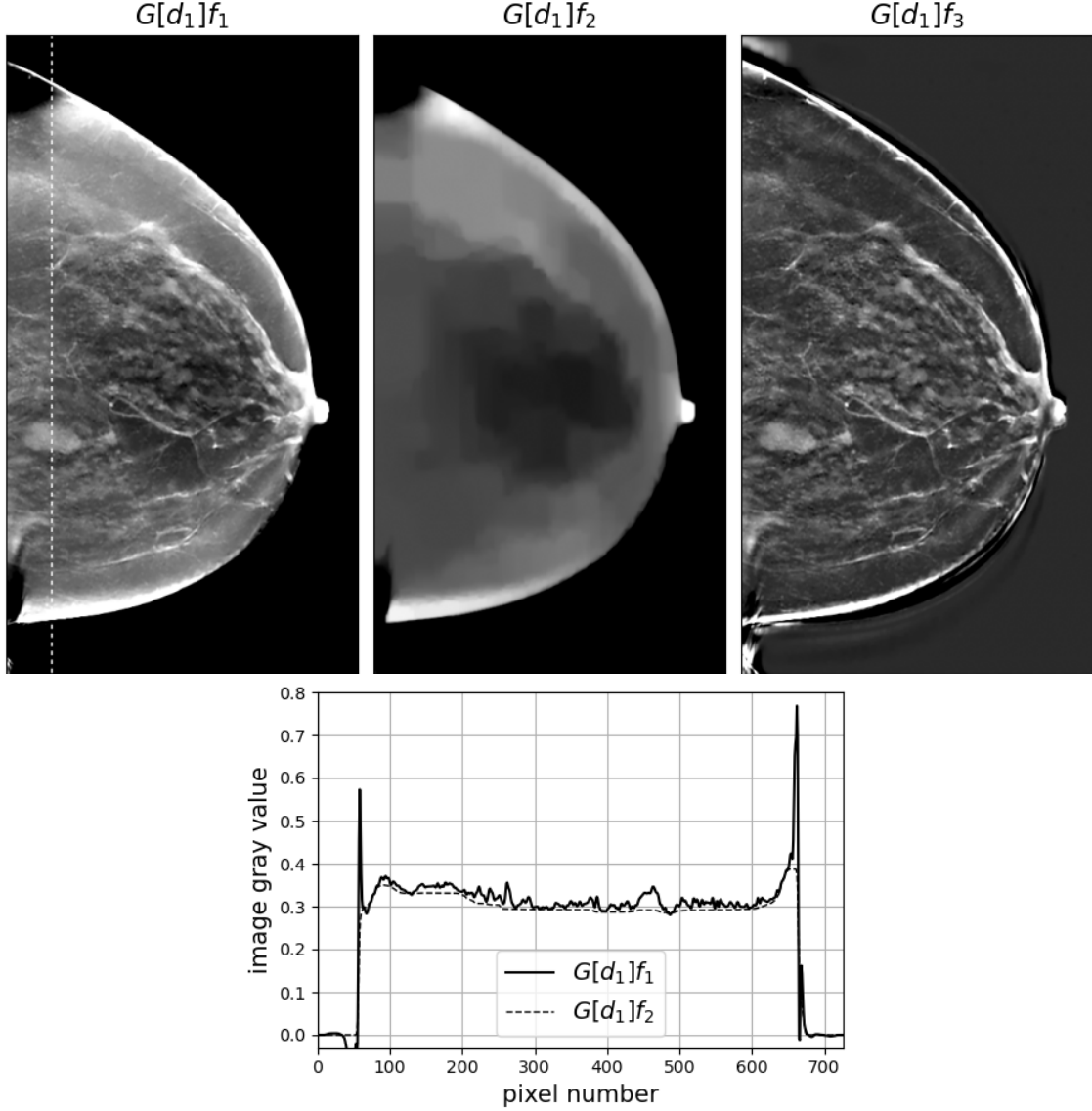


Fig 1 In-plane slice images for the coupled lowres-DTV-LSQ alongside a profile plot demonstrating the effectiveness of f_2 at extracting the image background. The dashed line in the f_2 image indicates the location of the profile plot data.

195 image reconstruction is carried out on a low-resolution voxel grid with voxels of size 340x340x680
 196 microns cubed, and the projection data are then down-sampled to a pixel grid with pixels of size
 197 340x340 microns squared.

The regularization parameter settings with which coupled lowres-DTV-LSQ is solved are

$$\alpha_y = \alpha_a = \alpha_b = \alpha_1 = 1/9; \quad \alpha_x = 5/9,$$

which are the same values used to perform image reconstruction on wide-angle DBT in Ref. 10, and the coupling parameter for the difference image f_3 is set to

$$\alpha_3 = 0.1,$$

which was tuned to a value that provided the best visual fit to the image background. The data RMSE constraints, also set by visual inspection, are both

$$\epsilon_1 = \epsilon_2 = 0.015.$$

These data RMSE constraints are the most critical parameters of coupled lowres-DTV-LSQ as they are the primary controls on the image regularization. The looser, i.e. larger, these values are, the more regularized f_1 and f_2 become. The design goal for this image reconstruction algorithm is to obtain the lowest image RMSE possible. From our simulations, the lowest image RMSE is obtained for the lowest value of ϵ_1 that still permits enough regularization that the speckle from noise is removed. For the present results, this is checked visually. Finally, the algorithm parameters for coupled lowres-DTV-LSQ are set to

$$\gamma = 5, \quad \rho = 1.75, \quad \text{and} \quad \beta = 100;$$

198 these parameters are explained in Sec. 6. Results for coupled lowres-DTV-LSQ are shown in
 199 Fig. 1, where an in-plane slice image is shown that contains a suspicious mass near the chest wall.
 200 The line profile and difference image show how f_2 is able to model the image background without
 201 following clinically relevant features such as the mass, vessels, or fibroglandular tissue, while f_2

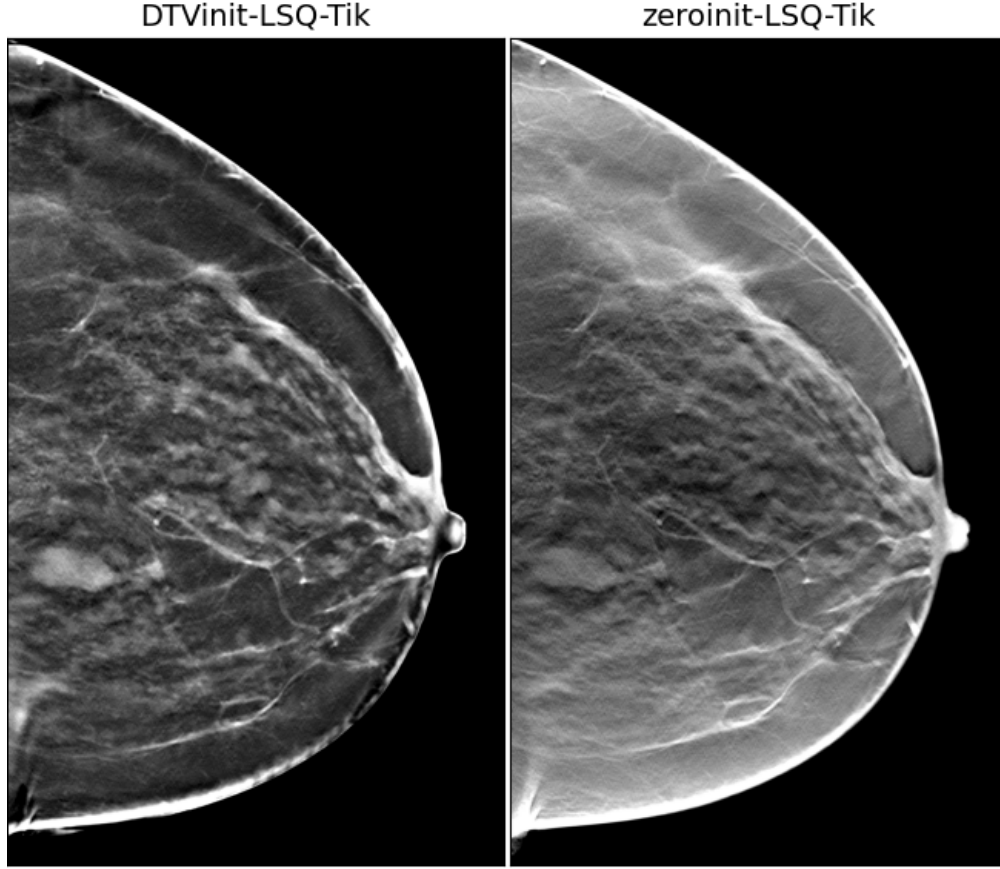


Fig 2 In-plane images of DTVinit-LSQ-Tik (left) and zeroinit-LSQ-Tik (right) for the same slice shown in Fig. 1.

is able to account for background low-frequency artifacts.

In order to use the results of coupled lowres-DTV-LSQ to form a full resolution DBT image, we follow the strategy of Ref. 10 and use f_1 to initialize a high-resolution Tikhonov-regularized least-squares (DTVinit-LSQ-Tik) image reconstruction

$$\arg \min_h \left\{ \frac{1}{2} \|R[c](XG[d]h - g)\|_2^2 + \alpha_{\text{Tik}} \frac{1}{2} \|G[d]h - h_0\|_2^2 \right\}, \quad (3)$$

where the high-resolution image h is composed of voxels of size 85x85x340 microns cubed; Ref. 10 explains how to form the initial image h_0 from the low-resolution image f_1 ; and the gaussian blurring width for DTVinit-LSQ-Tik is chosen to be the same as the voxel dimensions. The

algorithm consists of steepest descent steps on the LSQ-Tik objective function and the iteration is truncated at 10 steps. The parameter α_{Tik} controls the image regularization, and for this study this parameter is determined to be $\alpha_{\text{Tik}} = 0.1$ by visual appearance just for demonstration purposes. But it is recommended to determine α_{Tik} by objective task-based means as taught in FIS,¹ and demonstrated in Ref. 13.

In order to form the final image for display, the background image f_2 is upsampled to the high-resolution background image h_2 . A support image, h_{sup} , indicating voxels inside and outside of the breast with 1 and 0, respectively, is found by thresholding h_2 at a gray level of 0.1 cm^{-1} . A mean background gray value, b , is found by averaging h_2 in the breast support, i.e. where $h_{\text{sup}} = 1$. The image for display is then formed by

$$h_{\text{disp}} = G_z[d_z] ((h - h_2) + b \cdot h_{\text{sup}}),$$

where additional depth blurring with a one-dimensional gaussian kernel with width $d_z = 850$ microns (5 voxel lengths) is introduced for additional regularization. For comparison, the results of DTVinit-LSQ-Tik are compared with zeroinit-LSQ-Tik, which is found by the same algorithm except that the initial image h_0 has all voxel values set to zero. No background artifact removal for zeroinit-LSQ-Tik is performed. Example in-plane slice images are shown in Fig. 2. Because the background low-frequency artifacts are removed from DTVinit-LSQ-Tik, it is possible to display the image in a tighter grayscale window than the image for zeroinit-LSQ-Tik. Interestingly, the improved depth-resolution afforded by the DTV-LSQ initial image resolves the tangle of glandular tissue in the outer breast (image top) and improved in-plane contrast highlights the mass at the chest wall.

5 Conclusion

We have taken the opportunity of this special issue honoring Dr. Barrett to report on the latest developments on our work in digital breast tomosynthesis image reconstruction. The goal of the algorithm design is to improve depth resolution, in-plane contrast, and to remove non-uniformity artifacts. While the results from the shown clinical case show promise, it is clear that more studies are needed to demonstrate improved utility. Much of the future effort in this direction will involve carrying out image quality studies promoted by Dr. Barrett; clinical tasks sensitive to image reconstruction algorithms and parameters, e.g. detection of subtle signals such as microcalcifications, as we have done in Ref. 13, 14, or spiculations on the margin of a suspicious mass, will be formulated and employed to assess the algorithms in an objective manner.

The contributions Dr. Barrett has made permeate modern image science. The impact of his work goes far beyond the usual metrics of academic productivity, i.e. number of citations or successful grant applications. His work has been so foundational that it often goes uncited because it is part of the bedrock of the field. Thus we are grateful for the opportunity to spell out, in this work on DBT, the many ways that his contributions have influenced our work.

6 Appendix: PDHG implementation of Eq. (2)

This appendix shows how to fit Eq. (2) into the generic PDHG form

$$\min_x \{F(Kx) + G(x)\}, \quad (4)$$

where the F and G functions are convex and possibly non-smooth; K is a linear transforms; and x is the unknown vector. With a multiterm objective function such as the one presented in Eq. (2) the

function F ends up representing a sum of convex functions and K is formed by stacking multiple matrix blocks, which are the linear transforms inside of each of the objective terms. As shown in Sec. 2 of Ref. 10, we make the following associations

$$x = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{13} \end{bmatrix} = Kx, \quad \hat{F}(y) = \sum_{i=1}^{13} \hat{F}_i(y_i) = \sum_{i=1}^{13} F_i(y_i \|K_i\|_2 / \nu_i),$$

$$K = \begin{bmatrix} \nu_1 K_1 / \|K_1\|_2 & \cdot & \cdot \\ \nu_2 K_2 / \|K_2\|_2 & \cdot & \cdot \\ \vdots & \vdots & \vdots \\ \nu_6 K_6 / \|K_6\|_2 & \cdot & \cdot \\ \cdot & \nu_7 K_7 / \|K_7\|_2 & \cdot \\ \cdot & \nu_8 K_8 / \|K_8\|_2 & \cdot \\ \vdots & \vdots & \vdots \\ \cdot & \nu_{12} K_{12} / \|K_{12}\|_2 & \cdot \\ \cdot & \cdot & \nu_{13} K_{13} / \|K_{13}\|_2 \end{bmatrix},$$

where $\|K_i\|_2$ is the matrix norm, i.e. largest singular value, of the matrix K_i and ν_i are matrix scaling parameters that can be optimized for PDHG convergence efficiency. The function and

matrix block assignments are as follows

$$\begin{aligned}
F_1(y_1) &= \delta(y_1 \mid \|y_1 - R[c]g\|_2 \leq \epsilon_1 \cdot \sqrt{N_{\text{data}}}), \quad K_1 = R[c]XG[d_1] \\
F_7(y_7) &= \delta(y_7 \mid \|y_7 - R[c]G_{\text{det}}[d_d]g\|_2 \leq \epsilon_7 \cdot \sqrt{N_{\text{data}}}), \quad K_7 = R[c]XG[d_2], \\
F_2(y_2) &= \alpha_x \|y_2\|_1, \quad K_2 = \partial_x, \quad F_8(y_8) = \alpha_x \|y_8\|_1, \quad K_8 = \partial_x, \\
F_3(y_3) &= \alpha_y \|y_3\|_1, \quad K_3 = \partial_y, \quad F_9(y_9) = \alpha_y \|y_9\|_1, \quad K_9 = \partial_y, \\
F_4(y_4) &= \alpha_a \|y_4\|_1, \quad K_4 = \partial_a, \quad F_{10}(y_{10}) = \alpha_a \|y_{10}\|_1, \quad K_{10} = \partial_a, \\
F_5(y_5) &= \alpha_b \|y_5\|_1, \quad K_5 = \partial_b, \quad F_{11}(y_{11}) = \alpha_b \|y_{11}\|_1, \quad K_{11} = \partial_b, \\
F_6(y_6) &= \alpha_1 \|y_6\|_1, \quad K_6 = I, \quad F_{12}(y_{12}) = \alpha_1 \|y_{12}\|_1, \quad K_{12} = I, \\
F_{13}(y_{13}) &= \alpha_3 \|y_{13}\|_1, \quad K_{13} = I, \\
G(x) &= \delta([f_1, f_2, f_3] \mid f_1 - f_2 - f_3 = 0),
\end{aligned}$$

where the indicator function $\delta(x \mid C)$ is 0 (∞) if x does (does not) satisfy the equality or inequality C ; the oblique directional derivatives denoted with a and b are a combination of partials in the x and z direction

$$\partial_\theta = \cos \theta \partial_x + \sin \theta \partial_z,$$

241 where a and b denote the directions $\theta = -25^\circ$ and 25° , respectively.

Algorithm 1 PDHG update steps for solving Eq. (4). The normalized matrices $\hat{K}_i = K_i / \|K_i\|_2$ are used for conciseness.

- 1: $x^{(k+1)} \leftarrow (I + \tau \partial G)^{-1} \left(x^{(k)} - \tau \left(\sum_i \nu_i \hat{K}_i^\top \lambda_i^{(k)} \right) \right)$
 - 2: $\tilde{x} = 2x^{(k+1)} - x^{(k)}$
 - 3: $\lambda_i^{(k+1)} \leftarrow (I + \sigma_i \partial \hat{F}_i^*)^{-1} \left(\lambda_i^{(k)} + \sigma_i \nu_i \hat{K}_i \tilde{x} \right) \quad \forall i$
-

As explained in Ref. 10, the algorithm for solving Eq. (4) involves three update steps, which are repeated in Algorithm 1 for completeness. Lines 1 and 3 of this algorithm involve possibly unfamiliar notation which is fully explained in Ref. 10; the reason why the update steps are given is to specify the complete set of step-size parameters. Line 1 of Algorithm 1 involves the scalar step-size τ and the matrix scaling parameters ν_i ; Line 3 involves the scalar step-sizes σ_i . These step-size parameters are chosen so that

$$\tau \left\| \sum_i \sigma_i \nu_i^2 \hat{K}_i^\top \hat{K}_i \right\|_2 = 1, \quad (5)$$

which is based on the convergence condition for PDHG. Because the index i runs through all 13 matrix blocks, there are a total of 27 parameters (τ , σ_i , and ν_i) to tune for optimizing the PDHG convergence rate. The condition in Eq. (5) reduces this number to 26 independent parameters, but tuning 26 parameters is still a daunting task.

In Ref. 10, a framework for X-ray tomography is proposed that simplifies the parameter selection. In that work, it is proposed to provide heuristic arguments for selecting step-size weight parameters, $\hat{w}_i$, and ratio parameters r_i , which are related to τ , σ_i , and ν_i by the equations

$$w_i = w \hat{w}_i = \tau \sigma_i \nu_i, \text{ and } r_i = \sigma_i / \tau,$$

where $\hat{w}_i$ are relative weights and the scalar magnitude w is determined by Eq. (5)

$$w = \left\| \sum_i \hat{w}_i \hat{K}_i^\top \hat{K}_i \right\|_2^{-1}. \quad (6)$$

In Ref. 10, it is proposed to set all step-size ratios equal to a single scalar β and to weight all linear

transforms K_i the same except for the matrix blocks involving the X-ray transform, which should have greater weighting γ . For Eq. (2) there are two matrix blocks, K_1 and K_7 , that involve the X-ray transform. Accordingly, we make the following assignments

$$\begin{aligned}\hat{w}_1 &= \hat{w}_7 = \gamma, \\ \hat{w}_i &= 1, \quad \forall i \neq 1 \text{ or } 7, \\ r_i &= \beta,\end{aligned}$$

where γ is selected in the interval $\gamma \in [1, \infty]$, and β is a positive real number. As a result, the search space for the step-size parameters is reduced from 26 dimensions down to 2, which is much more manageable. As is done in Ref. 10, the reduction in step-size parameter search space allows the use of an accelerated predictor-corrector PDHG algorithm that comes with an additional extrapolation parameters ρ . The particular instantiation of the PDHG algorithm calls for derivation of proximal operators involving the function terms $F_i(y_i)$ and $G(x)$ and the details on how to derive those steps are presented in Ref. 10.

Disclosures

The authors have no relevant financial interests in the manuscript and no other potential conflicts of interest to disclose.

Code, Data, and Materials Availability

There is no code or data in this work for sharing.

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- 2 In-plane images of DTVinit-LSQ-Tik (left) and zeroinit-LSQ-Tik (right) for the same slice shown in Fig. 1.